

A Note on “Schwartz’s Lemma” (a useful quantitative estimate for zero-testing)

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The following famous and useful result:

Lemma 1 *Suppose K is any field, $f \in K[x_1, \dots, x_n] \setminus \{0\}$ has degree d , and $S \in K$ is any subset of finite cardinality N . Then f vanishes at at most dN^{n-1} points of S^n .*

...is commonly known as *Schwartz’s Lemma* or the *Schwartz-Zippel Lemma*. We prove Lemma 1 below.

Lemma 1 originated in the study of the complexity of deciding whether (putative) multivariate polynomial identities are true, and first appeared in a paper of Jack Schwartz in a 1979 ISSAC (International Symposium on Symbolic and Algebraic Computation) proceedings. Important precursors, with bounds weaker in certain cases, were derived by De Millo and Lipton in 1977 and Zippel in 1979. More recently, in 1999, Noga Alon derived his *Combinatorial Nullstellensatz*, which allows more flexibility in the subsets of K^n one can usefully sample from.

Let $\#S$ denote the cardinality of S . An immediate consequence of the preceding lemma is the following probabilistic estimate:

Corollary 1 *Following the notation of Lemma 1 above, assume x is picked uniformly randomly from S^n , i.e., every point of S is equally likely as a coordinate of x . Then the probability that f vanishes at x is at most $\min\left\{1, \frac{dN^{n-1}}{\min\{\#S, N\}^n}\right\}$. In particular, if both d and $\#S$ are at least N , then the preceding upper bound simplifies to $\frac{d}{N}$. ■*

In particular, one can think of Corollary 1 as a discrete analogue of the fact that, for $K \in \{\mathbb{R}, \mathbb{C}\}$, endowing K with any continuous positive probability density implies that the probability of picking a root of f is zero.

Proof of Lemma 1: First note that the statement is trivial when $d \geq N$: $dN^{n-1} \geq N^n$, the latter quantity is the cardinality of S^n , and f certainly can’t vanish at more points of S^n than N^n . So we may assume $d < N$.

We then proceed by induction on n . The base case is $n=1$ and follows immediately from the Weak Fundamental Theorem of Algebra. So let us assume our result true for a particular fixed value of n , and try to prove the obvious extension for n incremented by 1.

In particular, let us now assume that $f \in K[x_1, \dots, x_{n+1}]$ and write f in the form $\sum_{i=0}^d c_i(x_1, \dots, x_n)x_{n+1}^i$. The case where f is constant, and thus not identically zero by our hypotheses, is trivial: f vanishes at exactly $0 \leq dN^n$ points of S^{n+1} . So we may assume $d \geq 1$ (in addition to $d < N$). Let i be the greatest j such that c_j is *not* identically zero. (Thanks to our current assumptions, we must have $i \geq 1$.) Any $\zeta = (\zeta_1, \dots, \zeta_{n+1}) \in S^{n+1}$ must then fall into one of two exclusive cases:

$$(I) \quad c_i(\zeta_1, \dots, \zeta_n) = 0.$$

$$(II) \quad c_i(\zeta_1, \dots, \zeta_n) \neq 0.$$

Clearly, $\deg c_i \leq d - i$. So by our Induction Hypothesis, there can be no more than $(d - i)N^{n-1}$ points of S^n at which c_i can vanish. Clearly then, there are no more than $(d - i)N^{n-1} \cdot N = (d - i)N^n$ points (in S^{n+1}) of Type (I).

Now if $c_i(\zeta_1, \dots, \zeta_n) \neq 0$, then $f(\zeta_1, \dots, \zeta_n, x_{n+1})$ has degree i as a polynomial in x_{n+1} and thus vanishes at no more than i choices of x_{n+1} . In other words, there are no more than iN^n points in S^{n+1} of Type (II).

Since $(d - i)N^n + iN^n = dN^n$ we are done. ■